

Sleep Disorder Analysis

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Contents

1	Project overview	2
2	Data Validation and Cleaning	2
3	Preliminary Data Analysis	4
3.1	Target Variable: Sleep.Disorder	4
3.2	Demographics: Age and Gender	6
3.3	Lifestyle Factors: Stress Level, Physical Activity Level, and Daily Steps	7
3.4	Biometrics: Sleep Duration, Quality of Sleep	9
4	Questions	10
4.1	Is there an association between sleep disorders and BMI category?	10
4.1.1	Hypotheses	10
4.1.2	Statistical Method	10
4.1.3	Contingency Table	10
4.1.4	Chi-Square test	11
4.1.5	P-value	12
4.2	Do women sleep longer on average than men?	12
4.2.1	Hypotheses	12
4.2.2	Assumptions and checks	12
4.2.3	Test and results	14
4.2.4	Conclusion	15
4.3	Do stress levels differ between doctors, teachers, and engineers?	15
4.3.1	Hypotheses	16
4.3.2	Assumptions and checks	16
4.3.3	Test and results	16
4.3.4	Conclusion	17
4.4	Can we predict sleep quality based on lifestyle habits and physiological indicators?	17
4.4.1	Hypotheses	17
4.4.2	Assumptions and checks	17
4.4.3	Test and results	17
4.4.4	Model Accuracy (R^2 and Adjusted R^2)	18
4.4.5	Model Performance: Actual vs. Predicted Sleep Quality (Scatter Plot)	18
4.4.6	Conclusion	18
5	Overall conclusions and next steps	20
5.1	Summary of the dataset	20
5.2	Key findings from the research questions	20
5.3	Practical interpretation	21
5.4	Limitations	21
5.5	Final conclusion	21

1 Project overview

This project analyzes the **Sleep Disorder Diagnosis Dataset** (`sleep_disorder_dataset.csv`) to understand the **occurrence of sleep disorders**—primarily **Insomnia** and **Sleep Apnea**—and how they relate to **demographic**, **lifestyle**, and **physiological** factors. The dataset contains **~374 observations** and **14 (15 after cleaning) variables** describing participants' sleep habits and outcomes (e.g., **Sleep.Duration**, **Quality.of.Sleep**), lifestyle indicators (e.g., **Physical.Activity.Level**, **Stress.Level**, **Daily.Steps**, **Caffeine.Intake**), and biometric measures (e.g., **BMI.Category**, **Blood.Pressure**, **Heart.Rate**).

The analysis is designed to provide both descriptive insight (how common each sleep disorder is in this sample) and inferential/statistical conclusions about which factors are associated with sleep disorders and sleep quality. Following the project guidelines, the work begins with **data validation** in RStudio to ensure the dataset is imported correctly and contains no suspicious or missing values that could bias results, then proceeds through exploratory visualization and hypothesis-driven statistical analysis.

Core research questions include: - Is there an association between **sleep disorder occurrence** and **BMI category**? - Do **women** sleep longer on average than **men**? - Are **stress levels** different between **doctors, teachers, and engineers**? - Can **sleep quality** be predicted from lifestyle habits and physiological indicators using a regression-based model?

Overall, the project aims to translate the dataset into actionable statistical findings that clarify which measurable factors most strongly relate to sleep disorders and overall sleep quality in the observed population.

2 Data Validation and Cleaning

Ensuring that the dataset is clean, consistent, and reliable is a crucial step before analysis. Data cleaning involves checking each column for missing values, wrong entries, or inconsistencies and correcting them. Numeric columns, such as age, sleep duration, quality of sleep, physical activity level, stress level, heart rate, daily sleep, and caffeine intake, were summarized to confirm that all values fall within a reasonable range and no impossible values or outliers exist. Categorical variables, including gender, occupation, BMI category, and sleep disorder, were checked for consistency and duplication. Blood pressure was a special case, where the format of it is “systolic/diastolic”. It was carefully handled and split into 2 variables: systolic and diastolic. Additionally, the first few rows of the dataset were visually inspected, and unique values were reviewed to validate that no formatting errors were present. This detailed cleaning and validation process ensures that the dataset is ready for accurate exploratory analysis and statistical testing.

Firstly, we import the dataset into the variable `sleep_data`.

```
sleep_data <- read.csv("sleep_disorder_dataset.csv")
```

We check the structure of the dataset by looking at column names and verifying if there are any missing values. We do that by summarizing all variables, which provides an overview of variable types, value ranges, category frequencies, and the presence of missing value.

```
summary(sleep_data)
```

##	Person.ID	Gender	Age	Occupation
##	Min. : 1.00	Length:374	Min. :27.00	Length:374
##	1st Qu.: 94.25	Class :character	1st Qu.:35.25	Class :character
##	Median :187.50	Mode :character	Median :43.00	Mode :character
##	Mean :187.50		Mean :42.18	
##	3rd Qu.:280.75		3rd Qu.:50.00	
##	Max. :374.00		Max. :59.00	
##	Sleep.Duration	Quality.of.Sleep	Physical.Activity.Level	Stress.Level
##	Min. :5.800	Min. : 3.700	Min. :30.00	Min. :3.000
##	1st Qu.:6.400	1st Qu.: 6.300	1st Qu.:45.00	1st Qu.:4.000
##	Median :7.200	Median : 7.300	Median :60.00	Median :5.000

```
## Mean :7.132 Mean : 7.325 Mean :59.17 Mean :5.385
## 3rd Qu.:7.800 3rd Qu.: 8.400 3rd Qu.:75.00 3rd Qu.:7.000
## Max. :8.500 Max. :10.000 Max. :90.00 Max. :8.000
## BMI.Category Blood.Pressure Heart.Rate Daily.Steps
## Length:374 Length:374 Min. :63.00 Min. : 3000
## Class :character Class :character 1st Qu.:67.00 1st Qu.: 5600
## Mode :character Mode :character Median :70.00 Median : 7000
## Mean :70.16 Mean : 6817
## 3rd Qu.:72.00 3rd Qu.: 8000
## Max. :88.00 Max. :10000
## Sleep.Disorder Caffeine.Intake
## Length:374 Min. : 10
## Class :character 1st Qu.: 80
## Mode :character Median :120
## Mean :127
## 3rd Qu.:160
## Max. :340
```

The summary indicates that there are no missing values in the dataset and that all numerical variables, except blood pressure, which is stored in a combined format, fall within reasonable and expected ranges.

Before continuing the analysis, the **Blood Pressure** column requires special attention.

We inspect the first few values.

```
head(sleep_data$Blood.Pressure)
```

```
## [1] "126/83" "125/80" "125/80" "140/90" "140/90" "140/90"
```

```
unique(sleep_data$Blood.Pressure)
```

```
## [1] "126/83" "125/80" "140/90" "120/80" "132/87" "130/86" "117/76" "118/76"
## [9] "128/85" "131/86" "128/84" "115/75" "135/88" "129/84" "130/85" "115/78"
## [17] "119/77" "121/79" "125/82" "135/90" "122/80" "142/92" "140/95" "139/91"
## [25] "118/75"
```

Notice the form of values is systolic/diastolic.

We split into Systolic and Diastolic columns.

```
bp_split <- strsplit(as.character(sleep_data$Blood.Pressure),"/")
bp_matrix <- do.call(rbind, bp_split)
sleep_data$Systolic <- as.numeric(bp_matrix[,1])
sleep_data$Diastolic <- as.numeric(bp_matrix[,2])
```

After resolving this issue, the analysis proceeds with the examination of categorical variables.

Categorical variables were examined to verify the validity and consistency of category labels and to identify potential duplicates or misspellings.

```
categorical_cols <- c("Gender", "Occupation", "BMI.Category", "Sleep.Disorder")
```

```
lapply(sleep_data[categorical_cols], table)
```

```
## $Gender
##
## Female Male
## 185 189
##
## $Occupation
```

```
##
##           Accountant           Doctor           Engineer
##           37             71             63
##           Lawyer           Manager           Nurse
##           47             1             73
## Sales Representative       Salesperson       Scientist
##           2             32             4
##   Software Engineer       Teacher
##           4             40
##
## $BMI.Category
##
##           Normal Normal Weight       Obese       Overweight
##           195           21           10           148
##
## $Sleep.Disorder
##
##   Insomnia       None Sleep Apnea
##           77           219           78
```

An inconsistency was detected in the BMI category, where the labels “Normal” and “Normal Weight” referred to the same group. We merge those 2.

```
sleep_data$BMI.Category[sleep_data$BMI.Category == "Normal Weight"] <- "Normal"
table(sleep_data$BMI.Category)
```

```
##
##   Normal       Obese Overweight
##   216           10           148
```

Everything is consistent now.

Finally, we save the cleaned dataset.

```
write.csv(sleep_data, "sleep_data_cleaned.csv", row.names = FALSE)
```

3 Preliminary Data Analysis

After cleaning, we will begin exploratory data analysis to understand the dataset’s characteristics.

We will first look at the distribution of individual variables to get a baseline.

3.1 Target Variable: Sleep.Disorder

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

library(scales)

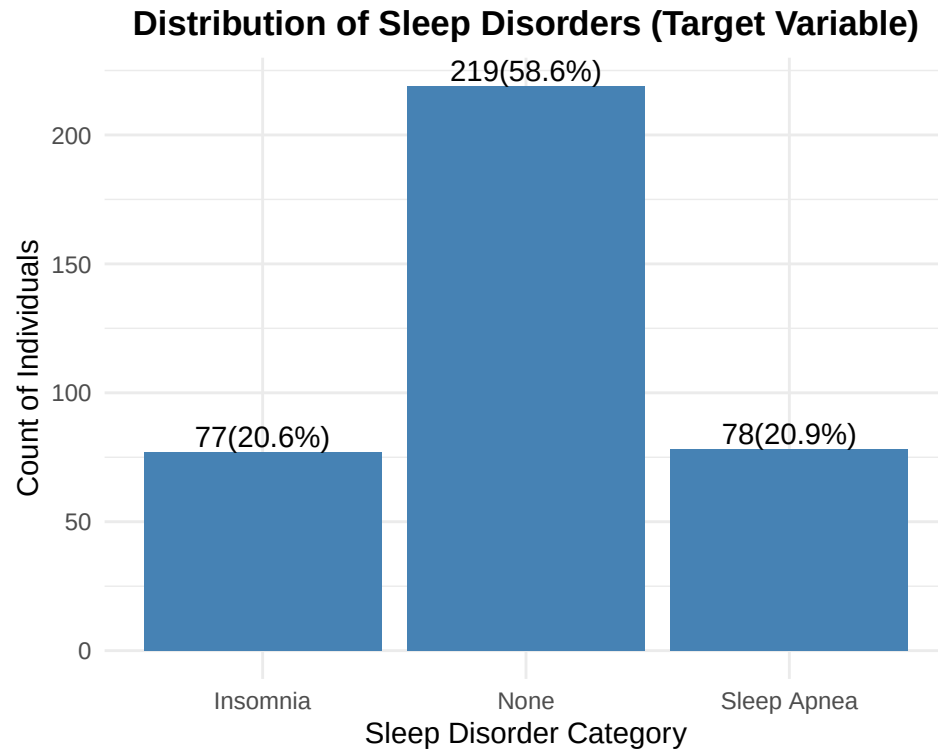
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor

df <- read.csv("sleep_data_cleaned.csv")

df <- df %>%
  mutate(Sleep.Disorder = replace_na(Sleep.Disorder, "None"))

# Generate the Bar Chart
ggplot(df, aes(x = Sleep.Disorder)) +
  geom_bar(fill = "steelblue") +
  # Add text labels with Count and Percentage
  geom_text(stat = "count",
            aes(label = paste0(after_stat(count), "(",
                               percent(after_stat(count) / sum(after_stat(count)), accuracy = 0.1), ")"),
            vjust = -0.25,
            size = 4) +
  # Apply formatting
  theme_minimal() +
  labs(title = "Distribution of Sleep Disorders (Target Variable)",
       x = "Sleep Disorder Category",
       y = "Count of Individuals") +
  theme(plot.title = element_text(hjust = 0.5, face = "bold"))

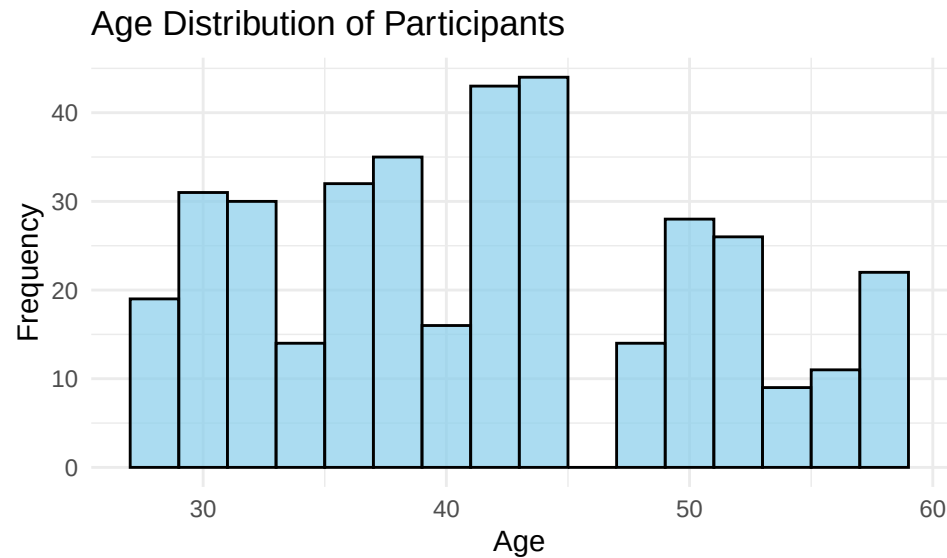
```



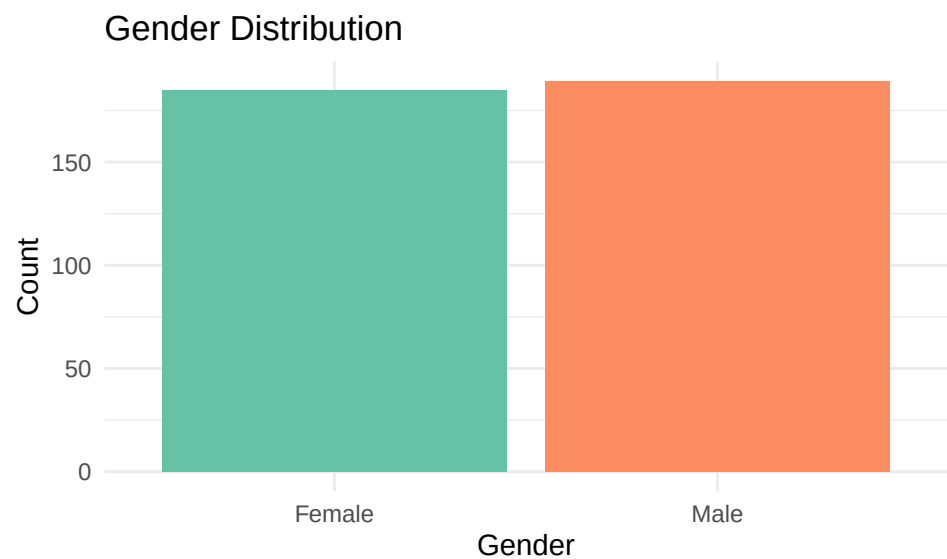
3.2 Demographics: Age and Gender

We will create a histogram for Age to see the age distribution of the participants and a bar chart for Gender to see the male/female split.

```
# Age Histogram
ggplot(sleep_data, aes(x = Age)) +
  geom_histogram(binwidth = 2, fill = "skyblue", color = "black", alpha = 0.7) +
  labs(title = "Age Distribution of Participants",
       x = "Age",
       y = "Frequency") +
  theme_minimal()
```



```
# Gender Bar Chart
ggplot(sleep_data, aes(x = Gender, fill = Gender)) +
  geom_bar() +
  labs(title = "Gender Distribution",
       x = "Gender",
       y = "Count") +
  theme_minimal() +
  scale_fill_brewer(palette = "Set2") +
  theme(legend.position = "none")
```



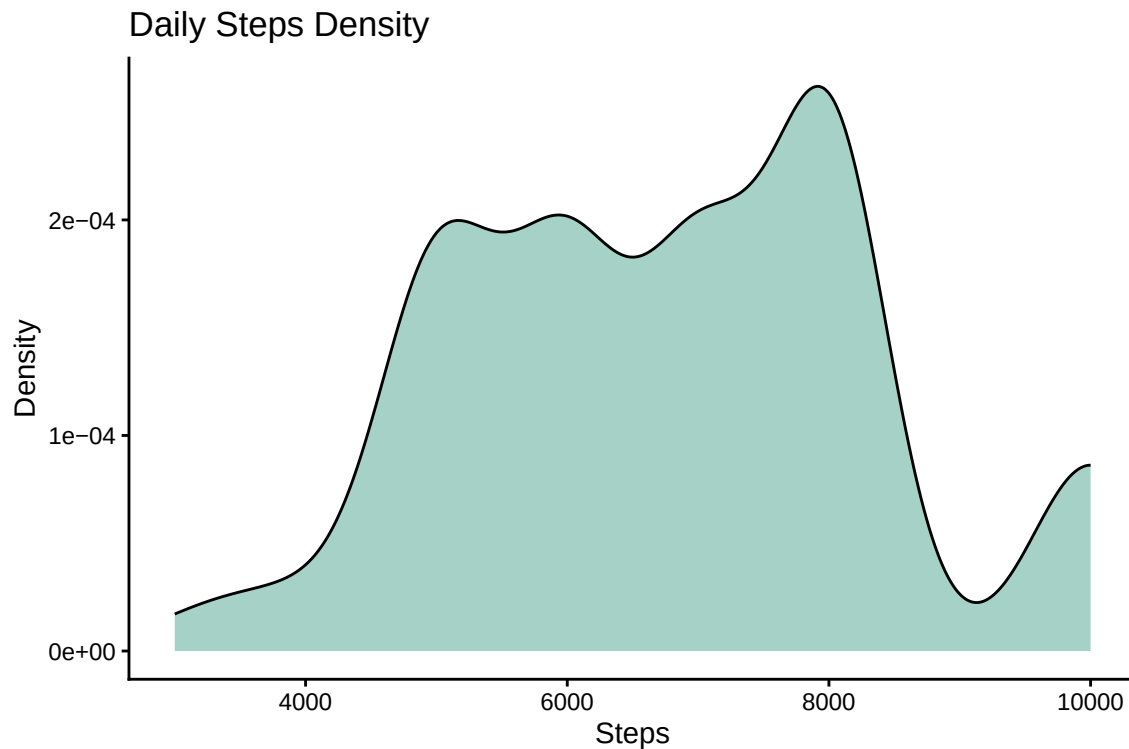
3.3 Lifestyle Factors: Stress Level, Physical Activity Level, and Daily Steps

Overall, participants appear **moderately active**. Most people record about **4,500-8,500 daily steps**, with a peak around **7,500-8,000**, and fewer reaching close to **10,000**.

```
library(readr)
library(ggplot2)
```

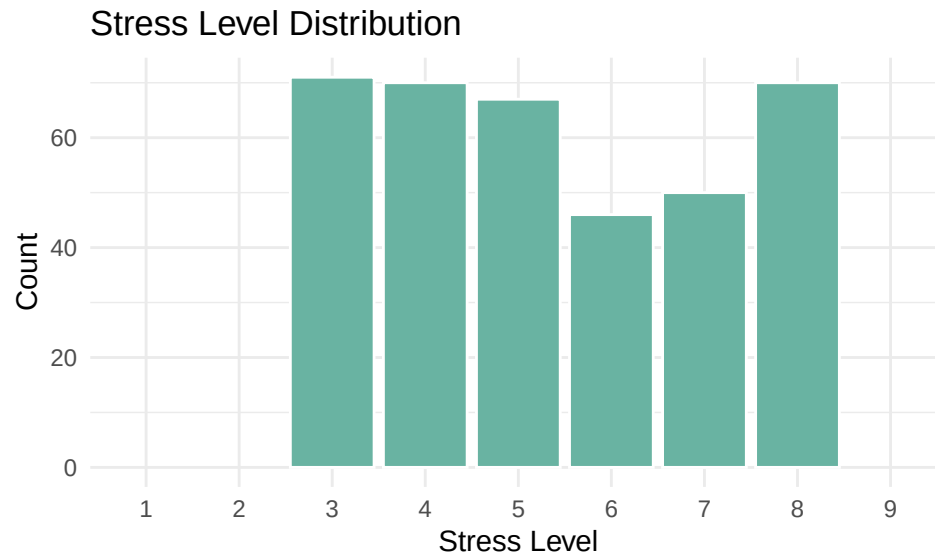
```
df <- read_csv("sleep_data_cleaned.csv")

ggplot(df, aes(x = Daily.Steps)) +
  geom_density(fill = "#69b3a2", alpha = 0.6) +
  theme_classic() +
  labs(title = "Daily Steps Density", x = "Steps", y = "Density")
```



Stress.Level is mostly concentrated in the mid-to-high range (about 3-8), indicating many participants report moderate to elevated stress.

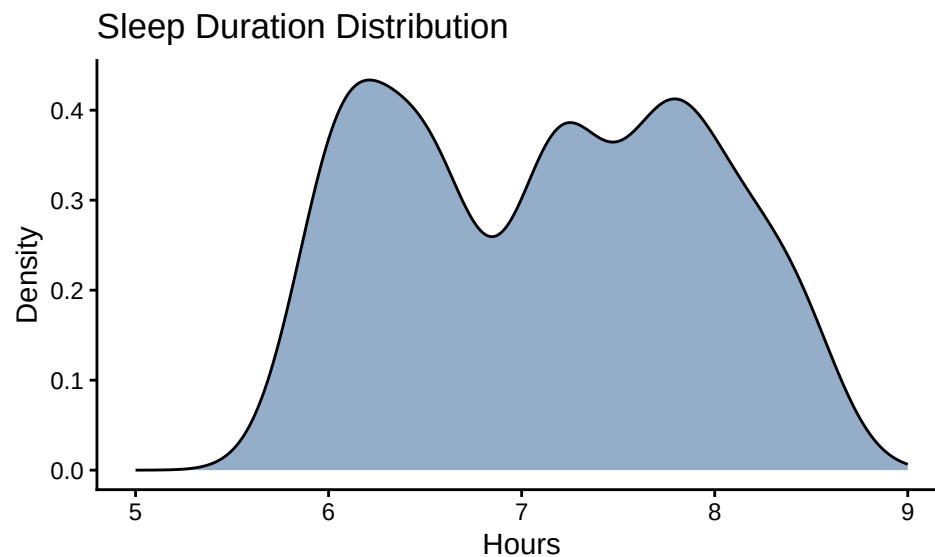
```
ggplot(df, aes(x = factor(Stress.Level, levels = 1:9))) +
  geom_bar(fill = "#69b3a2", color = "white") +
  scale_x_discrete(drop = FALSE) +
  theme_minimal() +
  labs(title = "Stress Level Distribution", x = "Stress Level", y = "Count")
```

3.4 Biometrics: Sleep Duration, Quality of Sleep

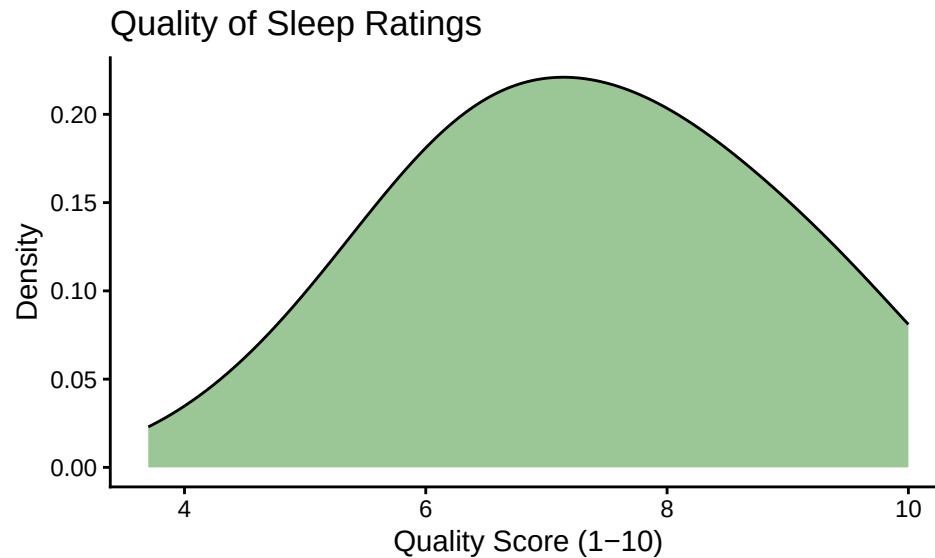
Most participants sleep about 6 to 8.5 hours, with clear clustering around common sleep lengths.

```
ggplot(df, aes(x = Sleep.Duration)) +
  geom_density(fill = "#4E79A7", alpha = 0.6) +
  xlim(5, 9) +
  theme_classic() +
  labs(title = "Sleep Duration Distribution", x = "Hours", y = "Density")
```



Sleep quality ratings are mostly mid-to-high, with the highest density around 7-8 on the 1-10 scale.

```
ggplot(df, aes(x = Quality.of.Sleep)) +
  geom_density(fill = "#59A14F", alpha = 0.6, adjust = 2) +
  theme_classic() +
  labs(title = "Quality of Sleep Ratings", x = "Quality Score (1-10)", y = "Density")
```



4 Questions

4.1 Is there an association between sleep disorders and BMI category?

Both variables are categorical: - **BMI.Category** (Underweight, Normal, Overweight, Obese) - **Sleep.Disorder** (None, Insomnia, Sleep Apnea)

Because we are analyzing the relationship between two categorical variables, we use a **Chi-Square Test of Independence**.

4.1.1 Hypotheses

- **Null Hypothesis (H_0):**
BMI category and sleep disorder are independent.
Knowing a person's BMI category gives no information about their sleep disorder.
- **Alternative Hypothesis (H_1):**
BMI category and sleep disorder are associated.
Knowing a person's BMI category gives information about their sleep disorder.

4.1.2 Statistical Method

The Chi-Square Test of Independence compares: - the **observed frequencies** of sleep disorders within each BMI category - the **expected frequencies** assuming the two variables are independent

Large differences between observed and expected counts indicate a possible association.

4.1.3 Contingency Table

```
bmi_sleep_table <- table(df$BMI.Category, df$Sleep.Disorder)
bmi_sleep_table
```

```
##
##           Insomnia None Sleep Apnea
## Normal           9 200           7
## Obese             4   0           6
## Overweight       64 19          65
```

```
chi_test <- chisq.test(bmi_sleep_table)
```

```
## Warning in chisq.test(bmi_sleep_table): Chi-squared approximation may be
## incorrect
```

```
expected_counts <- chi_test$expected
print(expected_counts)
```

```
##
##           Insomnia      None Sleep Apnea
## Normal      44.470588 126.481283  45.048128
## Obese        2.058824   5.855615   2.085561
## Overweight  30.470588  86.663102  30.866310
```

```
if (any(expected_counts < 5)) {
  print("Warning: Small expected frequencies detected")
}
```

```
## [1] "Warning: Small expected frequencies detected"
```

Since we have very small counts (less than 5) in some cells, the Chi-Square test's p-value calculation can become unreliable. Let's merge Overweight and Obese into a single one

```
df$BMI.Category.Merged <- as.character(df$BMI.Category)
df$BMI.Category.Merged[df$BMI.Category.Merged %in% c("Overweight", "Obese")] <- "Overweight/Obese"
df$BMI.Category.Merged <- as.factor(df$BMI.Category.Merged)
bmi_sleep_table_merged <- table(df$BMI.Category.Merged, df$Sleep.Disorder)
```

```
print("---- Observed Frequencies ----")
```

```
## [1] "---- Observed Frequencies ----"
```

```
bmi_sleep_table_merged
```

```
##
##           Insomnia None Sleep Apnea
## Normal           9  200           7
## Overweight/Obese  68  19           71
```

```
print("---- Expected Frequencies ----")
```

```
## [1] "---- Expected Frequencies ----"
```

```
chi_test <- chisq.test(bmi_sleep_table_merged)
print(chi_test$expected)
```

```
##
##           Insomnia      None Sleep Apnea
## Normal      44.47059 126.48128  45.04813
## Overweight/Obese 32.52941  92.51872  32.95187
```

4.1.4 Chi-Square test

```
chi_test <- chisq.test(bmi_sleep_table_merged)
```

```
chi_test
```

```
##
## Pearson's Chi-squared test
```

```
##
## data:  bmi_sleep_table_merged
## X-squared = 244.19, df = 2, p-value < 2.2e-16
```

4.1.5 P-value

```
chi_test$p.value
```

```
## [1] 9.425451e-54
```

```
if (chi_test$p.value < 0.05) {
  cat("Conclusion: The p-value is less than 0.05. We reject the null hypothesis and conclude that there is a significant difference between the two groups.")
} else {
  cat("Conclusion: The p-value is greater than or equal to 0.05. We fail to reject the null hypothesis and conclude that there is no significant difference between the two groups.")
}
```

```
## Conclusion: The p-value is less than 0.05. We reject the null hypothesis and conclude that there is a significant difference between the two groups.
```

4.2 Do women sleep longer on average than men?

To investigate whether women sleep longer than men, **sleep duration** was compared between two gender groups

4.2.1 Hypotheses

Let μ_1 be the mean sleep duration for women and μ_2 the mean sleep duration for men.

- **Null hypothesis (H_0):** $\mu_1 = \mu_2$
- **Alternative hypothesis (H_1):** $\mu_1 > \mu_2$

This is a one-tailed test because we specifically want to know whether women sleep *more* (than men).

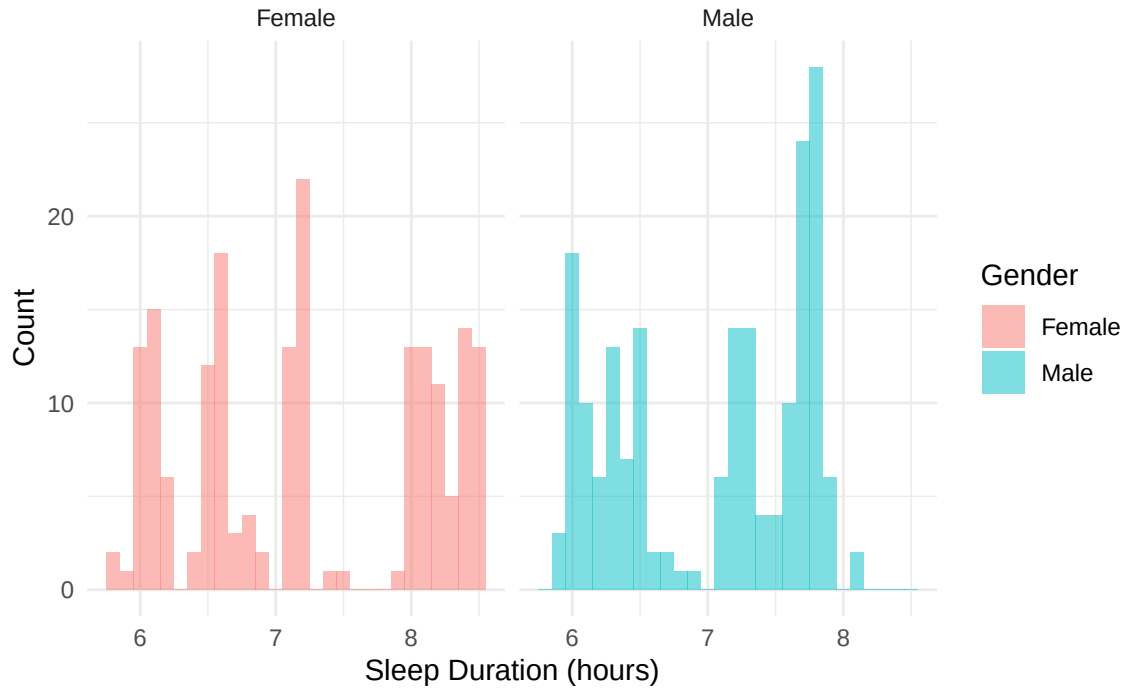
We set the **significance level** at $\alpha = 0.05$.

4.2.2 Assumptions and checks

Parametric tests, like the t-test, assume that the variable of interest is approximately normally distributed within each group. To assess whether this assumption holds for sleep duration, we first examined **histograms** of sleep duration separately for women and men.

```
ggplot(df, aes(x = Sleep.Duration, fill = Gender)) +
  geom_histogram(alpha = 0.5, position = "identity", binwidth = 0.1) +
  facet_wrap(~Gender) +
  theme_minimal() +
  labs(title = "Histogram of Sleep Duration by Gender",
       x = "Sleep Duration (hours)",
       y = "Count")
```

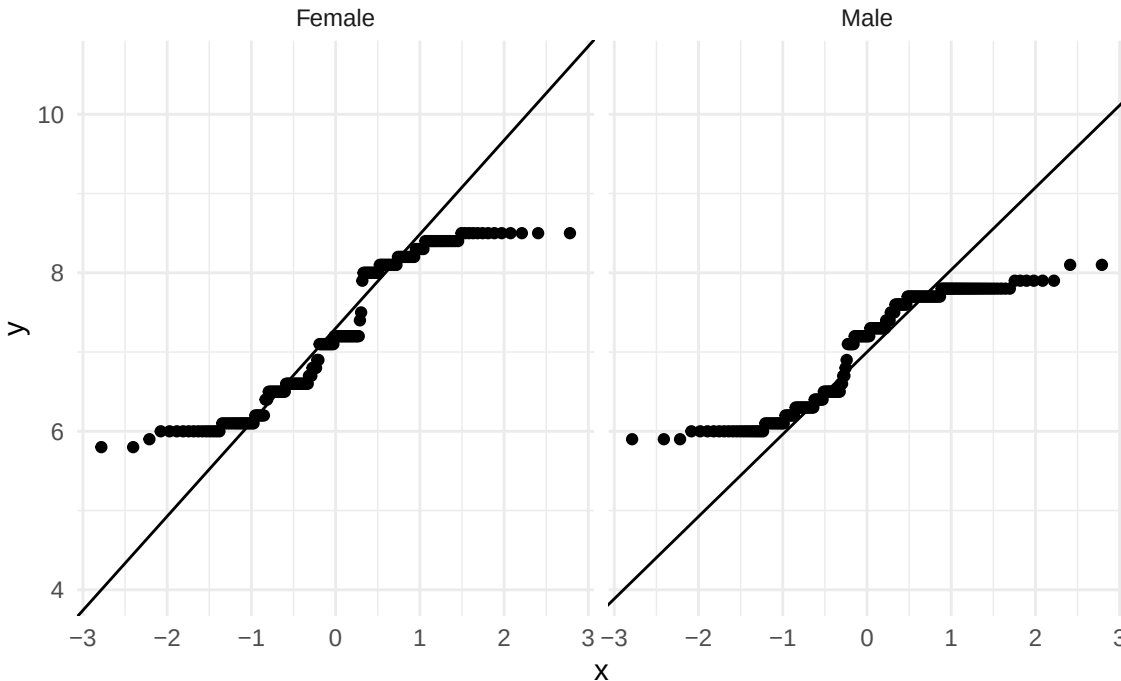
Histogram of Sleep Duration by Gender



From the histograms, sleep duration does not appear perfectly normally distributed within each gender. However, to assess normality more formally, we will examine **QQ plots** for both women and men.

```
ggplot(df, aes(sample = Sleep.Duration)) +  
  stat_qq() +  
  stat_qq_line() +  
  facet_wrap(~Gender) +  
  theme_minimal() +  
  labs(title = "QQ Plot of Sleep Duration by Gender")
```

QQ Plot of Sleep Duration by Gender



The QQ plots suggest that sleep duration does not follow a normal distribution for either gender, as points show substantial deviation from the reference line. To formally assess normality, the **Lilliefors test** was applied separately for women and men.

```
library(nortest)

lillie.test(df$Sleep.Duration[sleep_data$Gender == "Female"])

##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  df$Sleep.Duration[sleep_data$Gender == "Female"]
## D = 0.18242, p-value < 2.2e-16

lillie.test(df$Sleep.Duration[sleep_data$Gender == "Male"])

##
##  Lilliefors (Kolmogorov-Smirnov) normality test
##
## data:  df$Sleep.Duration[sleep_data$Gender == "Male"]
## D = 0.16223, p-value = 3.521e-13
```

Histograms and QQ plots suggested that sleep duration **does not** follow a normal distribution for either gender. The Lilliefors test confirmed this: for women, $D = 0.182$, $p < 2.2e-16$, and for men, $D = 0.162$, $p < 0.001$. Since the normality assumption is violated, a non-parametric test, such as the Mann-Whitney U test, is more appropriate for comparing sleep duration between genders.

4.2.3 Test and results

Mann-Whitney U test

```
wilcox.test(Sleep.Duration ~ Gender,
            data = df,
            alternative = "greater")
```

```
##
## Wilcoxon rank sum test with continuity correction
##
## data: Sleep.Duration by Gender
## W = 20036, p-value = 0.00722
## alternative hypothesis: true location shift is greater than 0
```

4.2.4 Conclusion

We performed a one-sided Mann-Whitney U test to compare sleep duration between women and men. The test returned a p-value of 0.007. Since this is less than the significance level of 0.05, we **reject** the null hypothesis and conclude that women sleep longer than men.

4.3 Do stress levels differ between doctors, teachers, and engineers?

To investigate whether **stress levels** differ by profession, **Stress.Level** was compared across three occupation groups: **Doctor**, **Teacher**, and **Engineer** (where *Software Engineer* was combined into **Engineer**).

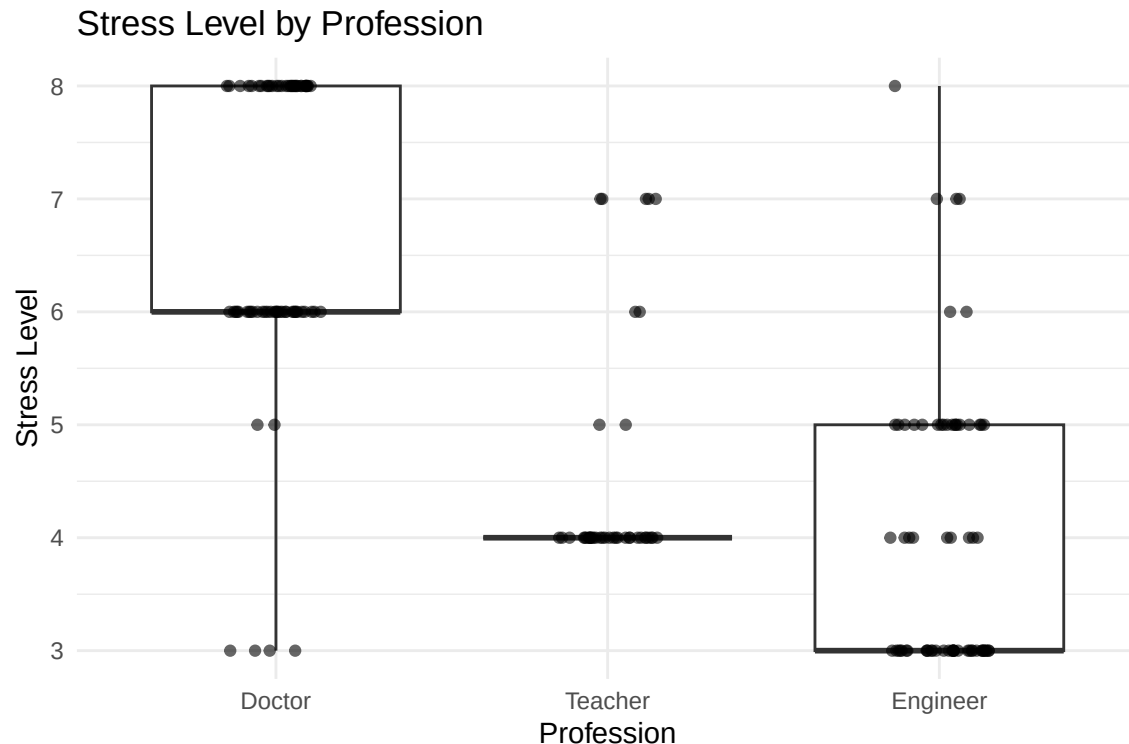
The boxplot suggests that **doctors tend to have higher stress levels** than teachers and engineers, while teachers and engineers appear more similar, though there is still some overlap between all groups.

```
library(dplyr)

df <- read_csv("sleep_data_cleaned.csv")

# Keep relevant occupations and merge Software Engineer into Engineer
dat <- df %>%
  filter(Occupation %in% c("Doctor", "Teacher", "Engineer", "Software Engineer")) %>%
  mutate(
    Group = if_else(Occupation == "Software Engineer", "Engineer", Occupation),
    Group = factor(Group, levels = c("Doctor", "Teacher", "Engineer"))
  )

ggplot(dat, aes(x = Group, y = Stress.Level)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.6) +
  theme_minimal() +
  labs(
    title = "Stress Level by Profession",
    x = "Profession",
    y = "Stress Level"
  )
```



4.3.1 Hypotheses

Let μ_1 be the mean stress level for doctors, μ_2 for teachers, and μ_3 for engineers. - H_0 : $\mu_1 = \mu_2 = \mu_3$
- H_1 : At least one group mean is different. We set the significance level at $\alpha = 0.05$.

4.3.2 Assumptions and checks

A one-way ANOVA assumes (among other things) that group variances are equal. We checked **equality of variances** using **Bartlett's test**.

```
bart_res <- bartlett.test(Stress.Level ~ Group, data = dat)
bart_res
```

```
##
## Bartlett test of homogeneity of variances
##
## data: Stress.Level by Group
## Bartlett's K-squared = 3.1165, df = 2, p-value = 0.2105
```

Bartlett's test returned a p-value greater than 0.05 ($p = 0.2105$), so we **fail to reject** the null hypothesis of equal variances. This indicates the equal-variance assumption is **not violated**, and it is appropriate to proceed with a standard one-way ANOVA.

4.3.3 Test and results

One-way ANOVA

```
anova_fit <- aov(Stress.Level ~ Group, data = dat)
anova_out <- summary(anova_fit)
anova_out
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
```



```
## Group          2  279.0  139.51   87.54 <2e-16 ***
## Residuals     175  278.9    1.59
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The one-way ANOVA produced $F(2, 175) = 87.5428$ with $p = 4.473e-27$, indicating a statistically significant difference in mean stress levels among the three professions.

4.3.4 Conclusion

Since the ANOVA p-value is far below $\alpha = 0.05$, we reject the null hypothesis and conclude that **stress levels differ significantly** between doctors, teachers, and engineers in this dataset. Descriptively, doctors show the **highest average stress levels**, while teachers and engineers report lower average stress.

4.4 Can we predict sleep quality based on lifestyle habits and physiological indicators?

To determine whether sleep quality can be predicted by various lifestyle and physiological factors, we fitted a multiple linear regression model. We used `Quality.of.Sleep` as the dependent variable and a set of independent variables: lifestyle (e.g., Stress Level, BMI), physiological (e.g., Blood Pressure) indicators.

To evaluate the model's performance, we calculated both Multiple R-squared and Adjusted R-squared. While Multiple R-squared indicates the total proportion of variance explained by the model, we primarily relied on Adjusted R-squared to assess effectiveness, as it corrects for the number of predictors and prevents overestimation of the model's fit. Additionally, to identify which specific factors significantly influence sleep quality, we examined the t-statistics for each independent variable. A significant t-statistic (with a p-value < 0.05) indicates that the specific factor has a statistically significant linear relationship with sleep quality.

4.4.1 Hypotheses

In multiple linear regression, we test the significance of the coefficient β_j for each predictor j , holding all other variables constant.

- **Null hypothesis (H_0):** $\beta_j = 0$
(There is no linear relationship between predictor j and Sleep Quality; the variable has no effect).
- **Alternative hypothesis (H_1):** $\beta_j \neq 0$
(There is a significant linear relationship; the variable has a non-zero effect on Sleep Quality).

We perform this two-sided t-test for every predictor included in the model.

4.4.2 Assumptions and checks

Before interpreting the results, we performed two key checks to ensure the model is valid:

1. **Independence of predictors:** We checked that the variables (like blood pressure components and stress) are distinct enough and do not heavily overlap with each other (no critical multicollinearity).
2. **Sample Size:** Since we have a large number of observations ($n = 374$), the statistical tests used by the model are reliable and robust, even if the data isn't perfectly normally distributed.

4.4.3 Test and results

We fitted the linear model using `lm()` in R and examined the **t-statistics** and corresponding **p-values** for each coefficient from the model summary (`summary(model)`).

Significant findings:

1. **Stress Level:** The test returned a large negative t-statistic with a **p-value** < 0.001 .
2. **BMI Category (Overweight):** The test returned a negative t-statistic with a **p-value** $= 0.002$.
3. **Age:** The test returned a positive t-statistic with a **p-value** $= 0.002$.

At the **5% significance level** ($\alpha = 0.05$), the p-values for these factors are smaller than α . Therefore, we **reject the null hypothesis** (H_0) for Stress Level, BMI (Overweight), and Age, concluding that they significantly influence sleep quality.

Non-significant findings: For variables such as **Physical Activity**, **Heart Rate**, and **Sleep Duration**, the p-values were greater than 0.05. Thus, we **fail to reject the null hypothesis** for these factors, meaning there is insufficient evidence to claim they affect sleep quality in this specific multivariate context.

4.4.4 Model Accuracy (R^2 and Adjusted R^2)

To evaluate how well our model fits the data, we calculated two key metrics:

- **Multiple R-squared:** 0.57
- **Adjusted R-squared:** 0.56

What do these numbers mean? The **Adjusted R-squared of 0.56** indicates that our model explains approximately **56% of the variance** in sleep quality using the selected lifestyle and physiological factors. We focus on the *Adjusted* metric because it corrects for the number of predictors, providing a more conservative and accurate measure.

This is considered a **fairly good result**. While we capture more than half of what determines sleep quality, the remaining ~44% suggests that other unmeasured factors—such as environment (noise, temperature, mattress quality) or genetics—also play a significant role. Given that we only used basic lifestyle data, explaining 56% of the variation is a strong outcome.

4.4.5 Model Performance: Actual vs. Predicted Sleep Quality (Scatter Plot)

To visually assess the model's accuracy, we plotted the **Actual** sleep quality scores against the **Predicted** scores (*Plot 8*).

The plot includes a **black dashed line** representing perfect prediction ($y = x$) and a **red solid line** showing the model's trend. The data points generally cluster along the diagonal, confirming that the model correctly captures the main trend: better lifestyle factors correspond to higher predicted sleep quality.

However, the visible spread of points around the line reflects the **~44% unexplained variance**. This visual pattern is consistent with the Adjusted R^2 of 0.56, indicating that the model is reliable for predicting general trends but lacks precision for specific individual outliers.

4.4.6 Conclusion

There is **statistical evidence** that **Stress Level**, **BMI Category**, and **Age** are significant predictors of sleep quality. Specifically, lower stress, normal weight, and older age are associated with higher sleep scores. However, the model explains approximately **56% of the variance**, suggesting that while these lifestyle factors are crucial, other unmeasured individual differences also play a significant role.

R code for conducting the multiple linear regression analysis and drawing the prediction plot:

```
# Build Linear Regression Model
model <- lm(Quality.of.Sleep ~ Age + Gender + Physical.Activity.Level + Stress.Level +
            BMI.Category + Heart.Rate + Daily.Steps + Sleep.Duration + Systolic + Diastolic,
            data = df)

# Output Results
summ <- summary(model)

cat("Model Accuracy Metrics:\n")
```

```
## Model Accuracy Metrics:
```

```

cat("Multiple R-squared:", round(summ$r.squared, 3), "(Variance explained)\n")

## Multiple R-squared: 0.574 (Variance explained)
cat("Adjusted R-squared:", round(summ$adj.r.squared, 3), "(Adjusted for predictors)\n\n")

## Adjusted R-squared: 0.561 (Adjusted for predictors)
# Significant Factors
coeffs <- summary(model)$coefficients
significant_factors <- coeffs[coeffs[, 4] < 0.05, ]

print("Significant Factors affecting Sleep Quality (p-value < 0.05):")

## [1] "Significant Factors affecting Sleep Quality (p-value < 0.05):"
print(significant_factors)

##
##              Estimate Std. Error   t value    Pr(>|t|)
## (Intercept)    6.08661366  2.76604150   2.200478 2.840406e-02
## Age            0.05031237  0.01642538   3.063088 2.354674e-03
## Stress.Level   -0.46927672  0.08188435  -5.730970 2.104811e-08
## BMI.CategoryOverweight -0.80262961  0.26342602  -3.046888 2.481939e-03
# Visualization: Actual vs Predicted
df$Predicted_Sleep_Quality <- predict(model)

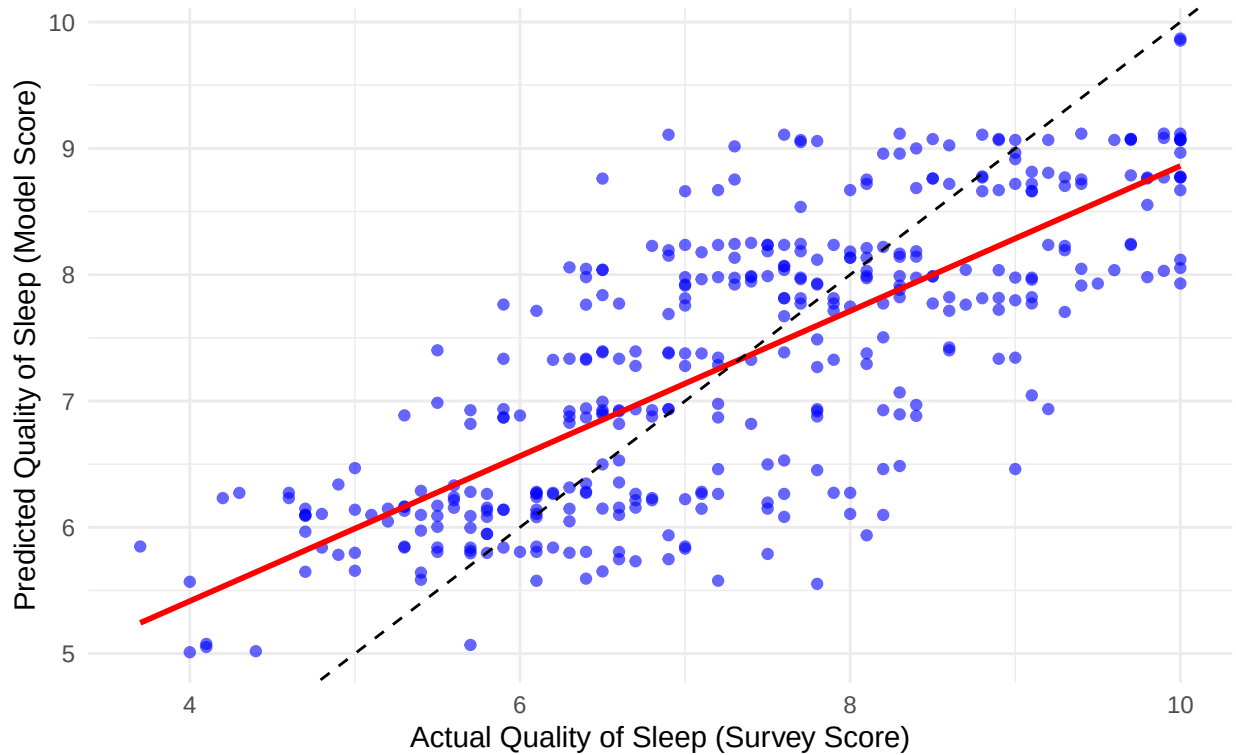
ggplot(df, aes(x = Quality.of.Sleep, y = Predicted_Sleep_Quality)) +
  geom_point(color = "blue", alpha = 0.6) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "black") +
  labs(
    title = "Model Performance: Actual vs Predicted Sleep Quality",
    x = "Actual Quality of Sleep (Survey Score)",
    y = "Predicted Quality of Sleep (Model Score)",
    subtitle = "Black dashed line = Perfect Prediction. Red line = Our Model trend."
  ) +
  theme_minimal()

## `geom_smooth()` using formula = 'y ~ x'

```

Model Performance: Actual vs Predicted Sleep Quality

Black dashed line = Perfect Prediction. Red line = Our Model trend.



5 Overall conclusions and next steps

5.1 Summary of the dataset

This analysis used **374 observations** with a nearly balanced gender split (**189 men, 185 women**). Participants were aged **27-59** (*mean* ≈ 42.2). Average sleep duration was ≈ 7.13 **hours** (range **5.8-8.5**), and average self-reported sleep quality was ≈ 7.32 (range **3.7-10**). Most participants recorded ≈ 6817 **daily steps** on average (range **3,000-10,000**), and stress was typically moderate (*median* = 5 on a 1-9 scale).

Regarding the target outcome, **58.6%** reported **no sleep disorder** (coded as “None”), while **Insomnia (20.6%)** and **Sleep Apnea (20.9%)** were both common in this sample.

5.2 Key findings from the research questions

1) Sleep disorder and BMI category are strongly associated.

After merging “Overweight” and “Obese” due to small expected counts, the Chi-square test showed an extremely strong relationship ($p < 0.001$). Practically, the pattern is very clear: in the **Normal BMI** group, most individuals fall in the “None” category, while the **Overweight/Obese** group contains the majority of **Insomnia** and **Sleep Apnea** cases.

2) Women sleep slightly longer than men on average.

Because sleep duration violated normality assumptions, a one-sided Mann-Whitney U test was used. The result ($p = 0.007$) supports the conclusion that women sleep longer than men in this dataset. However, the difference is modest in size: mean sleep duration is ≈ 7.23 **hours for women** versus ≈ 7.04 **hours for men** (about **12 minutes** difference on average).

3) Stress levels differ substantially by profession (Doctor vs Teacher vs Engineer).

ANOVA results indicate strong evidence of differences among the three occupation groups ($p < 0.001$). Descriptively, **doctors have the highest stress levels** (mean ≈ 6.73), while **engineers** (mean ≈ 4.01) and **teachers** (mean ≈ 4.53) are notably lower.

4) **Sleep quality is moderately predictable from the available lifestyle and physiological variables.** The multiple linear regression achieved $R^2 \approx 0.57$ and **Adjusted $R^2 \approx 0.56$** , meaning the model explains about **56%** of variability in sleep quality. The strongest and most consistent predictor is **Stress Level** (negative association). **Age** (positive association) and **BMI (Overweight vs Normal)** (negative association) also remain statistically significant once other variables are controlled. In contrast, variables like physical activity, heart rate, and sleep duration did not show independent significance in the multivariable model-likely because their effects overlap with other predictors or are not linear in this dataset.

5.3 Practical interpretation

Overall, the dataset suggests a coherent story: **higher stress and higher BMI align with poorer sleep outcomes**, both in terms of **sleep disorder prevalence** and **lower sleep quality scores**. While this analysis cannot establish causality, it supports targeting **stress reduction** and **healthy weight management** as plausible intervention points, and it highlights BMI category as a strong marker for identifying individuals at higher risk of insomnia or sleep apnea within this sample.

5.4 Limitations

Several limitations should be kept in mind when interpreting results:

- **Observational / cross-sectional design:** associations do not imply causation.
- **Self-reported measures:** sleep quality and related variables may contain reporting bias.
- **Category imbalance:** some categories are small (e.g., “Obese” is rare in this dataset), which required merging BMI groups and can reduce granularity.
- **Model assumptions and omitted variables:** the regression explains 56% of variance, but the remaining 44% likely reflects unmeasured factors such as mental health, environmental conditions (noise, light), medication use, alcohol intake, screen time, genetics, and work schedules.

5.5 Final conclusion

In conclusion, sleep disorders are common in this dataset (about **41%** combined insomnia/sleep apnea), and the strongest relationships observed are the links between **BMI category and sleep disorder**, and between **stress and sleep quality**. The regression model suggests that a relatively small set of measurable factors-especially stress level, BMI, and age-captures a meaningful portion of differences in sleep quality, but substantial unexplained variation remains, motivating future work with richer behavioral and environmental data.